WORKED SOLUTION

STEP I 1987, Question 4

Pure • Logarithms and Geometric Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q4

The key idea

Logarithms to awkward bases are best turned into natural logarithms, because then the base shows up as a simple factor. Every base here is a power of 2, and the exponents 2^n come down as multipliers. Once each term is written in terms of $\ln 2$, the series is just a geometric series in disguise.

Solution

Rewriting a general term

Fix $n \geq 0$ and let $c = \log_{2^{2^n}} \mathrm{e}$. By the definition in the hint, this means

$$(2^{2^n})^c = \mathrm{e}.$$

Taking natural logarithms of both sides (both are positive),

$$c \cdot 2^n \ln 2 = \ln \mathrm{e} = 1, \qquad \text{so} \qquad \log_{2^{2^n}} \mathrm{e} = \frac{1}{2^n \ln 2}.$$

As a check on the indexing, $n = 0, 1, 2$ give the bases 2, 4, 16, matching the first three terms, with values $\frac{1}{\ln 2}$, $\frac{1}{2\ln 2}$, $\frac{1}{4\ln 2}$.

? **Ask yourself** Where did the 2^n come from?

From the rule $\ln(a^k) = k \ln a$, used with $a = 2$ and $k = 2^n$: the base 2^{2^n} has natural logarithm $2^n \ln 2$. The "tower" of exponents is what makes each base the *square* of the one before, and squaring the base halves the logarithm.

Summing the series

The series is therefore

$$\sum_{n=0}^{\infty} (-1)^n \log_{2^{2^n}} \mathrm{e} = \frac{1}{\ln 2} \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n.$$

This is a geometric series with first term 1 and common ratio $r = -\frac{1}{2}$. Since $|r| < 1$ it converges, to $\dfrac{1}{1-r} = \dfrac{1}{1+\frac{1}{2}} = \dfrac{2}{3}$. Hence

$$\sum_{n=0}^{\infty} (-1)^n \log_{2^{2^n}} \mathrm{e} = \frac{2}{3\ln 2} = \frac{1}{\frac{3}{2}\ln 2} = \frac{1}{\ln(2^{3/2})} = \frac{1}{\ln(2\sqrt{2})},$$

using $\ln(a^k) = k \ln a$ again, and $2^{3/2} = 2 \cdot 2^{1/2} = 2\sqrt{2}$. This is the required sum.

Checks

Numerically, $\frac{1}{\ln(2\sqrt{2})} \approx 0.96180$. The partial sums of the series are 1.4427, 0.7213, 1.0820, 0.9017, 0.9919, 0.9468, …, swinging either side of this value with the gap halving each time, as an alternating geometric series should. Sixty terms agree with 0.961797 to machine precision.

Common slips

- Reading $\log_{2^{2^n}}$ as $\log_{2^{2n}}$: that would give bases 2, 16, 256, … and the wrong ratio $-\frac{1}{4}$. Check the indexing against the first few terms.
- Writing $\log_a \mathrm{e} = \ln a$ instead of $\dfrac{1}{\ln a}$. The change-of-base rule gives $\log_a b = \dfrac{\ln b}{\ln a}$, and here $\ln b = 1$.

Beyond STEP

Whether an infinite sum "has a value" at all is the central question of a first course in analysis. Geometric series are the basic benchmark: many convergence tests, such as the ratio test, work by comparing a series with a geometric one. Abbott's *Understanding Analysis* builds up this theory carefully and readably.

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